Sri Lanka Institute of Information Technology

Data warehousing and Business Intelligence

Assignment 2

WIDANAGAMAGE S S S – IT23641006

Year 03 Semester 01 - WE 02.01

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1.Introduction

This report documents the implementation completed for IT3021 Assignment 2. The work uses the data warehouse that was designed and loaded in Assignment 1 as the source for analytical processing and then extends the solution with SQL Server Analysis Services (SSAS), Excel-based OLAP analysis, and Power BI reporting.

The implemented solution is based on a retail sales domain. Three operational/master-data sources were integrated into a SQL Server staging database and then transformed into the Retail_DW2 warehouse. That warehouse was then used to build a multidimensional SSAS cube, an Excel workbook connected to the cube, and four Power BI reports.

2. Data source for Assignment 2

In this implementation, the source warehouse is Retail_DW2 in SQL Server that was created in Assignment 1. The warehouse stores denormalized dimension data and one central fact table in a star schema design so that it can support multidimensional analysis and reporting.

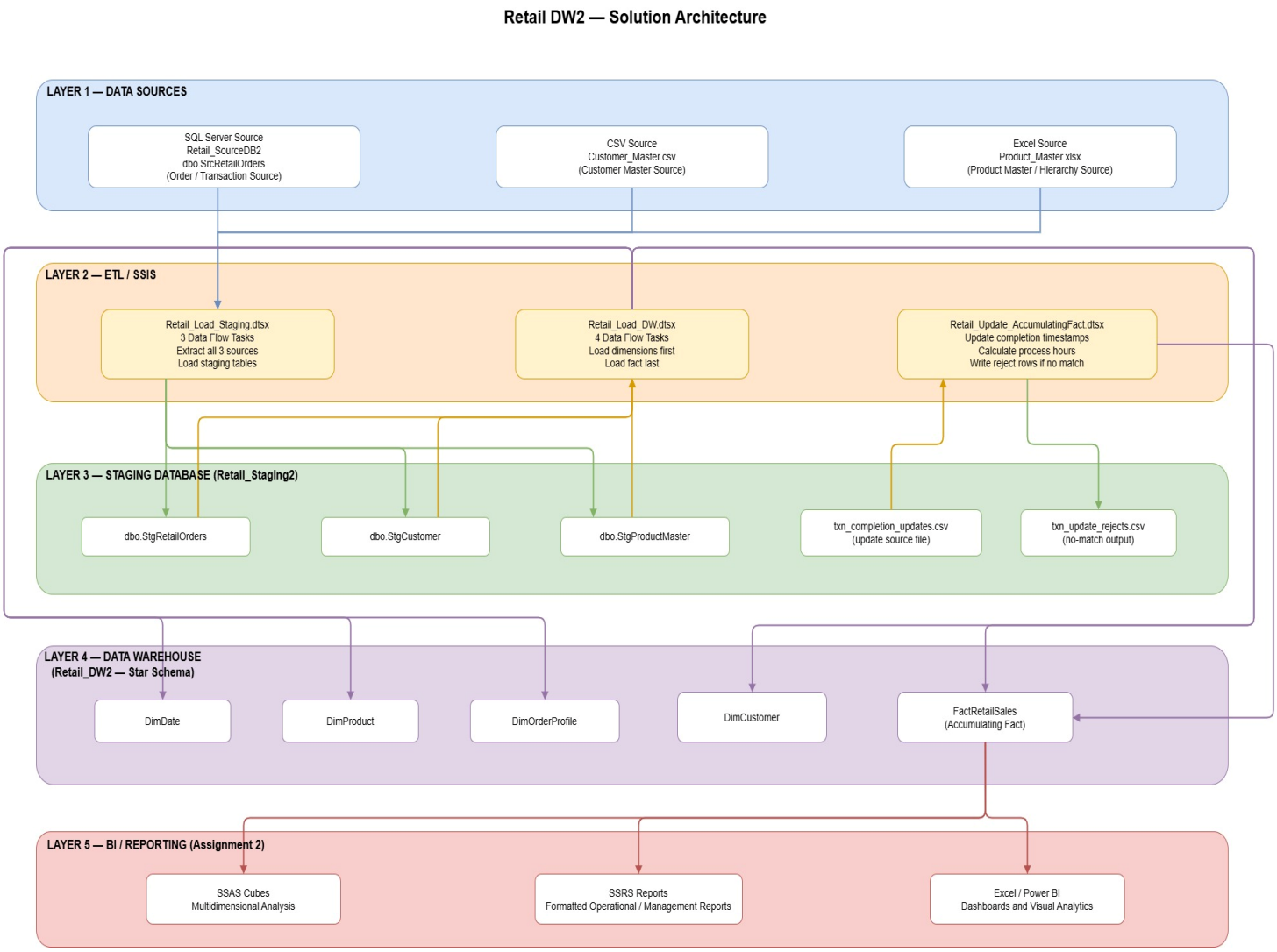
2.1 Source systems and staging layer

The retail data integration solution combines three source types. Transaction and order data comes from the SQL Server operational source Retail_SourceDB2.dbo.SrcRetailOrders. Customer master data comes from the flat file Customer_Master.csv. Product master and hierarchy data comes from the Excel file Product_Master.xlsx. These files first extracted into the staging database Retail_Staging2 through SSIS and then loaded into the data warehouse.

Source system summary

Layer	Component	Purpose
Operational source	Retail_SourceDB2.dbo.SrcRetailOrders	Primary transactional order source containing dates, quantities, amount values, and order profile attributes.
External master data	Customer_Master.csv	Customer profile source used to populate customer dimension attributes.
External master data	Product_Master.xlsx	Product master and hierarchy source used to populate product dimension attributes.
Staging database	Retail_Staging2	Landing zone used to decouple extraction from warehouse loading and support transformation logic.
Warehouse	Retail_DW2	Star schema warehouse used as the Assignment 2 data source.

Architecture Diagram



2.2 Data warehouse design (Retail_DW2)

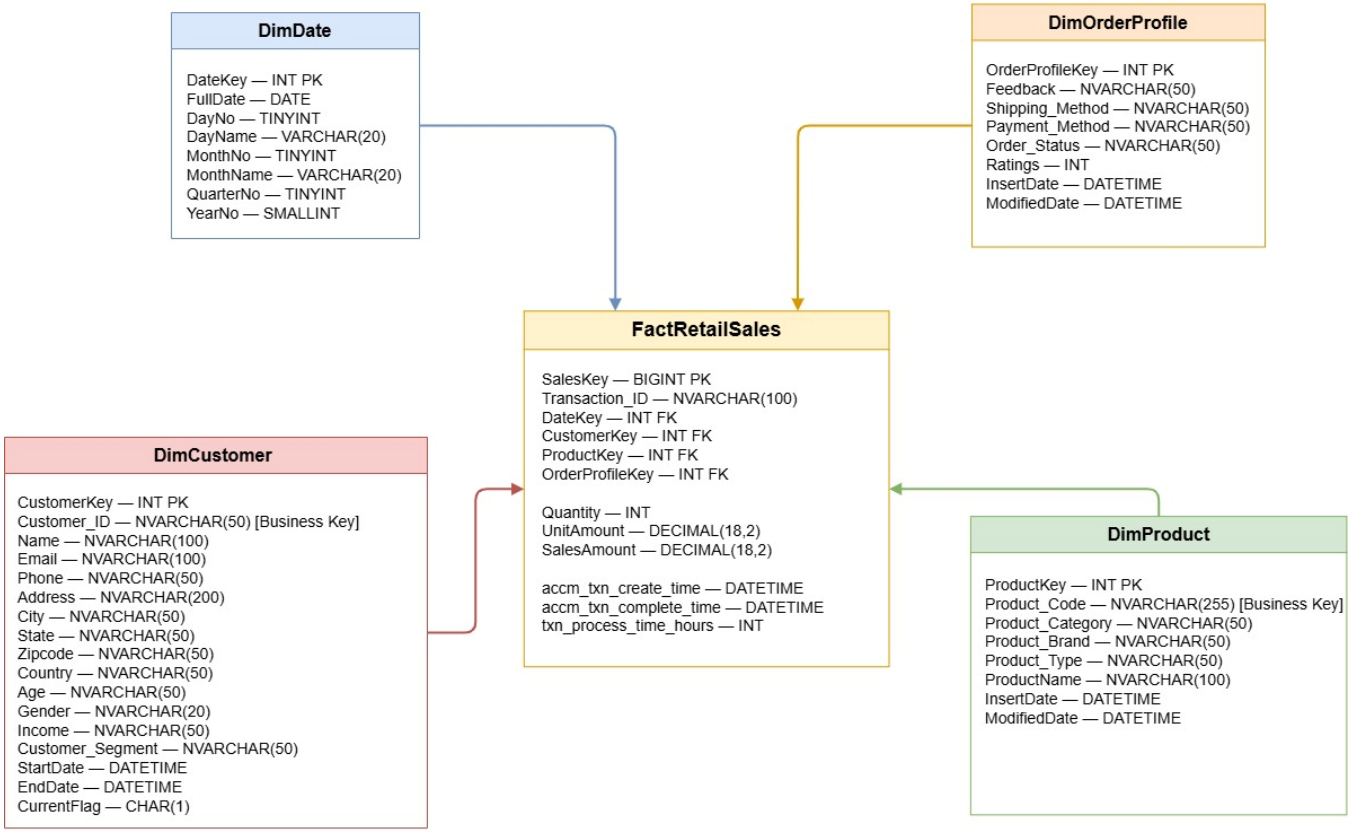
Retail_DW2 was implemented as a star schema. The central fact table is FactRetailSales. It is linked to four dimensions: DimDate, DimCustomer, DimProduct, and DimOrderProfile. All analysis is performed through the fact table.

Retail_DW2 warehouse objects

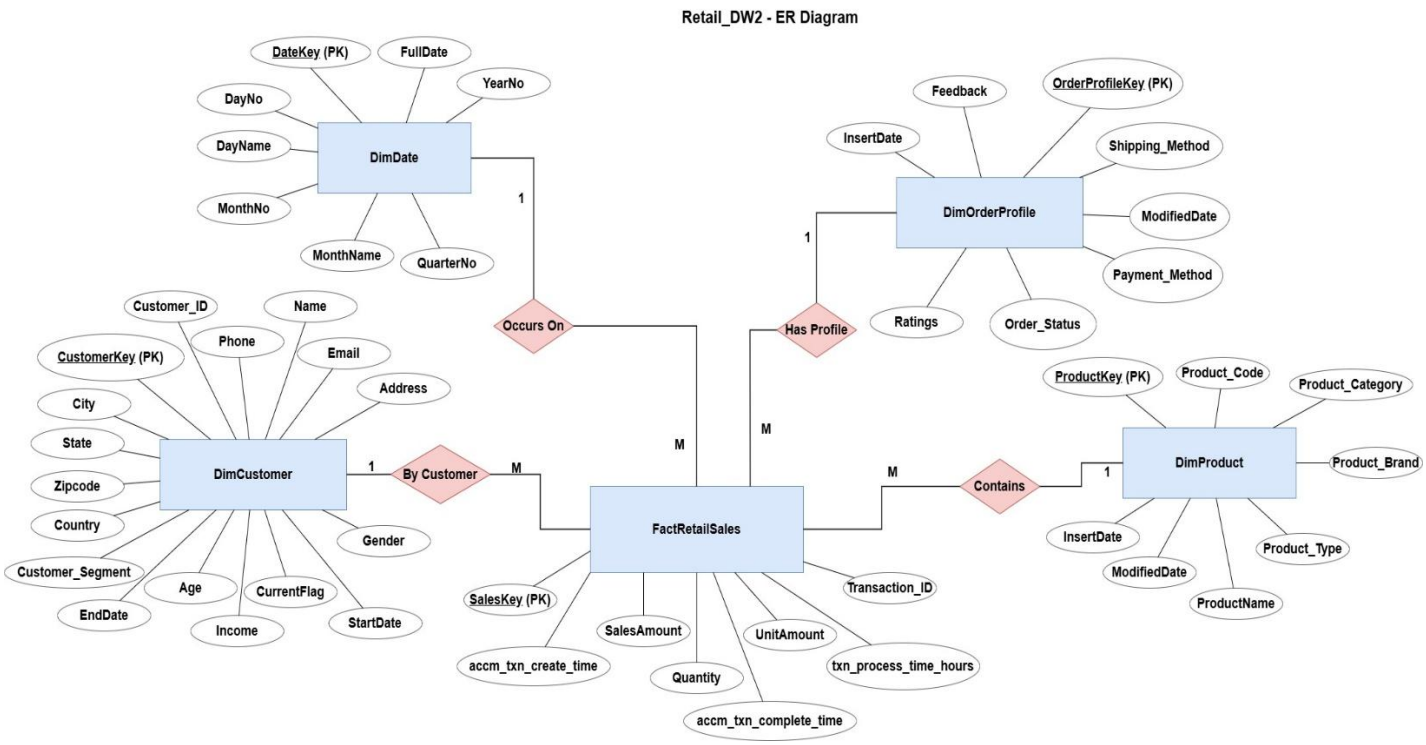
Table	Role in warehouse	Key design notes
DimDate	Date dimension	Stores FullDate, DayNo, DayName, MonthNo, MonthName, QuarterNo and YearNo.
DimCustomer	Customer dimension	Implemented StartDate, EndDate and CurrentFlag to preserve selected customer history.
DimProduct	Product dimension	Stores business key Product_Code plus product category, brand, type and product name.
DimOrderProfile	Order profile dimension	Stores Feedback, Shipping_Method, Payment_Method, Order_Status and Ratings.
FactRetailSales	Fact table	Stores one row per retail transaction with Quantity, UnitAmount, SalesAmount and accumulating-process timestamps.

Dimensional Model – Star Schema

Retail_DW2 — Star Schema Dimensional Model

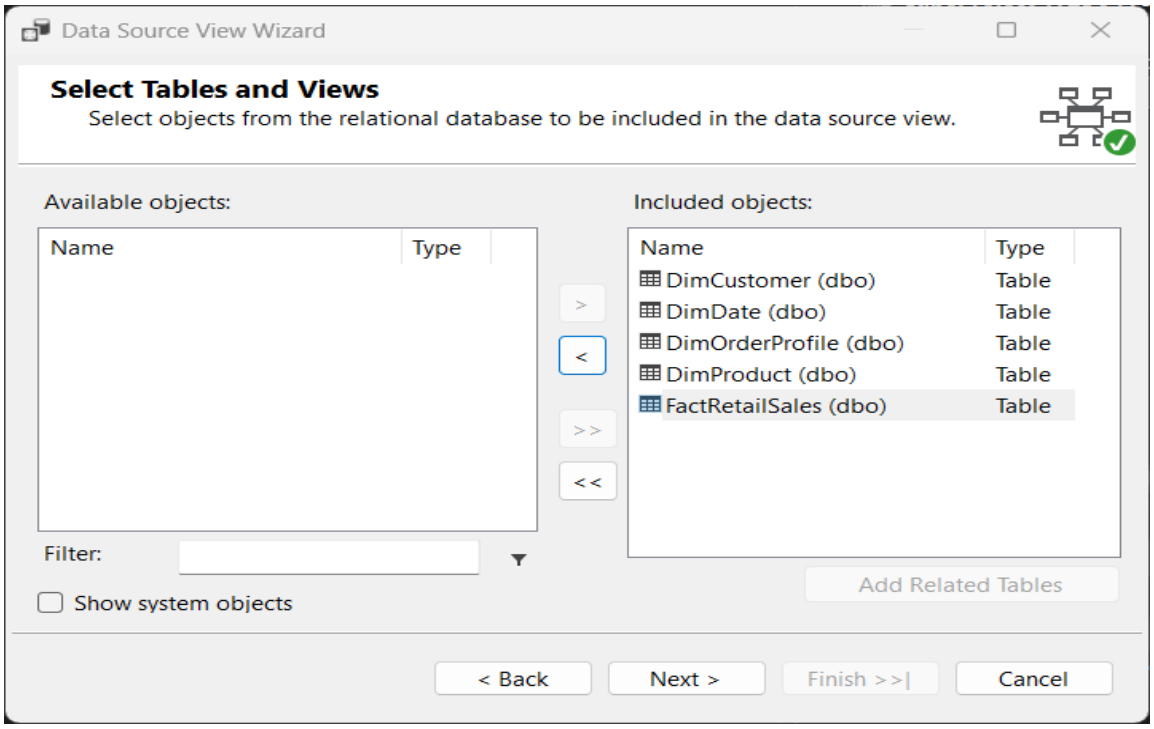


Entity Relationship Diagram



Screenshot evidence for Step 1

Data Source View table selection showing the dimension tables and FactRetailSales included for cube modelling.



3. SSAS cube implementation

3.1 Project configuration and deployment targets

The Analysis Services project was created as CubeProject_IT23641006. The relational SQL source used by the data source object was localhost\SASHEEN with database Retail_DW2. The deployment target for Analysis Services was SASHEENSRI\WIDANAGAMAGE, and the SSAS database name used during deployment was Retail_DW2_SSAS_IT23641006.

SSAS implementation configuration

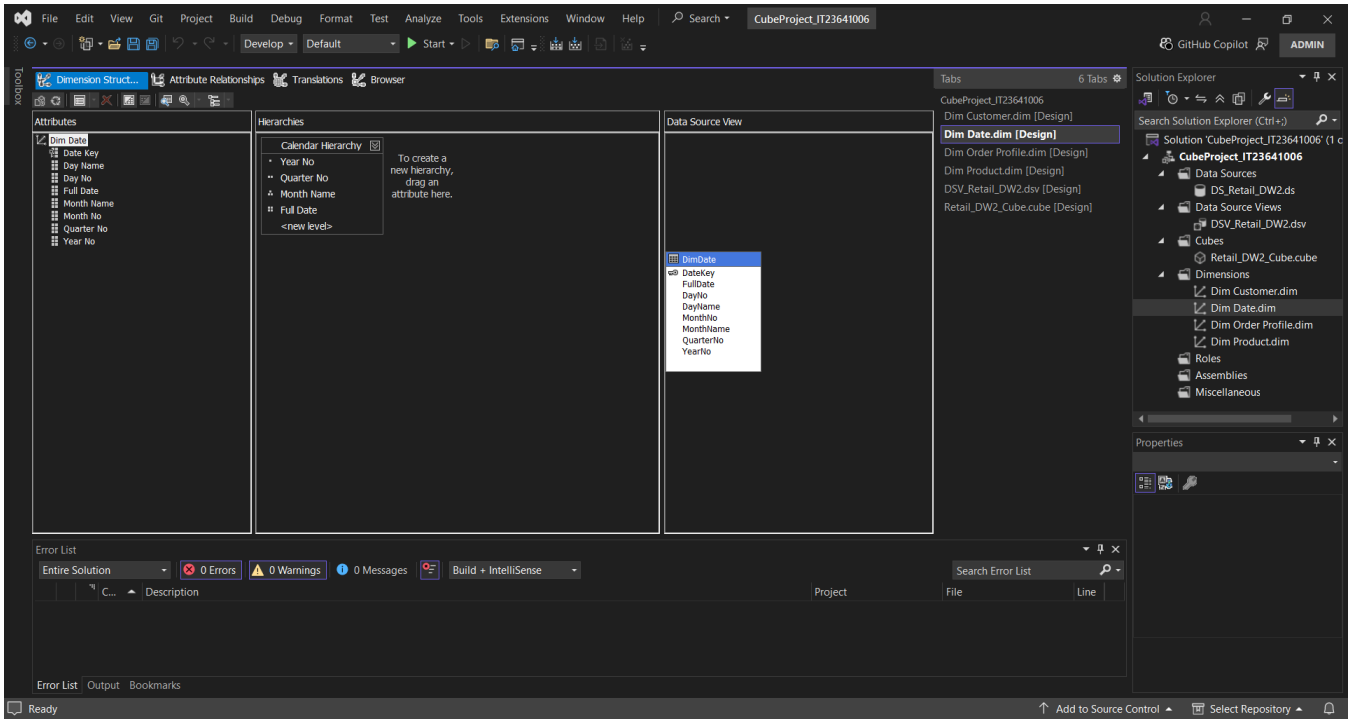
Project element	Value used
Visual Studio project	CubeProject_IT23641006
SSAS cube name	Retail_DW2_Cube
Data source name	DS_Retail_DW2
Data source view	DSV_Retail_DW2
Relational SQL source	localhost\SASHEEN / Retail_DW2
SSAS server	SASHEENSRI\WIDANAGAMAGE
SSAS database	Retail_DW2_SSAS_IT23641006

3.2 Data source, DSV, dimensions and hierarchies

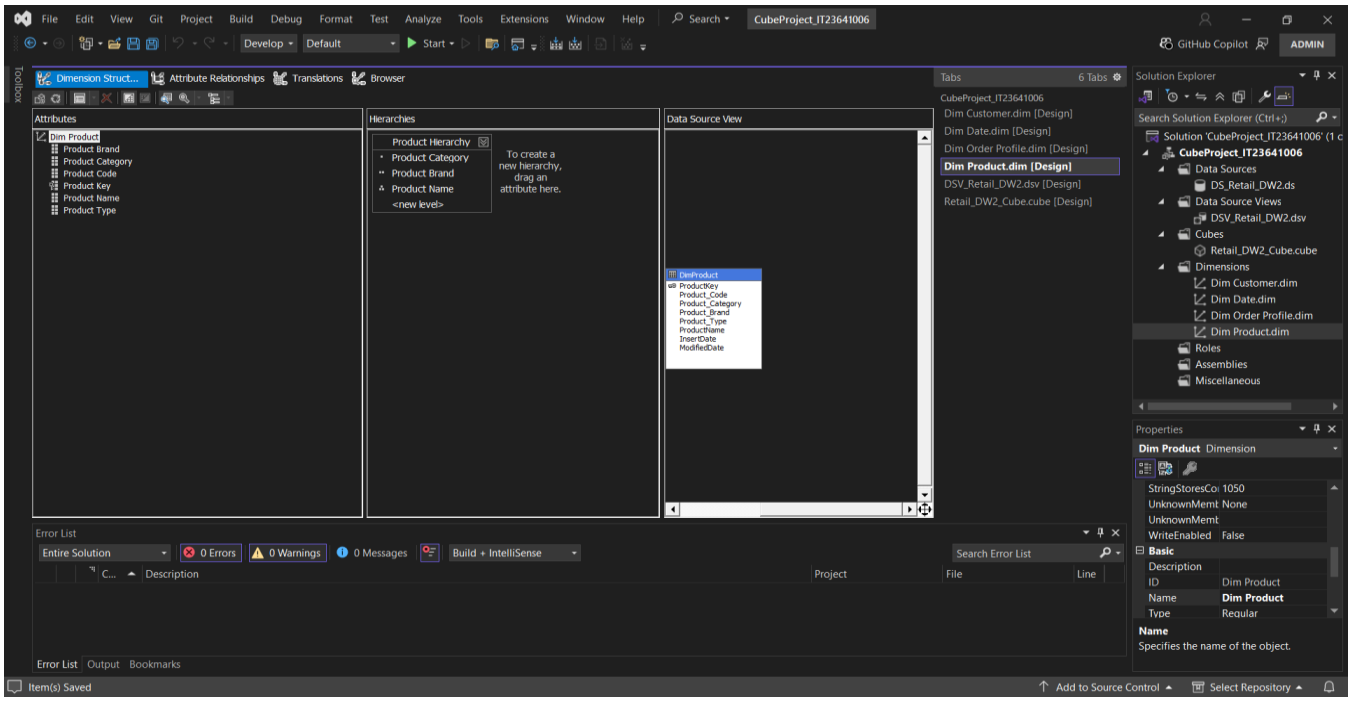
After creating the project, a relational data source was created for Retail_DW2. Then Data Source View was created using the warehouse tables: DimCustomer, DimDate, DimOrderProfile, DimProduct and FactRetailSales. Next four dimensions were created based on the dimension tables, and the fact table.

- Dim Customer: key column CustomerKey.
- Dim Date: key column DateKey.
- Dim Product: key column ProductKey.
- Dim Order Profile: key column OrderProfileKey.

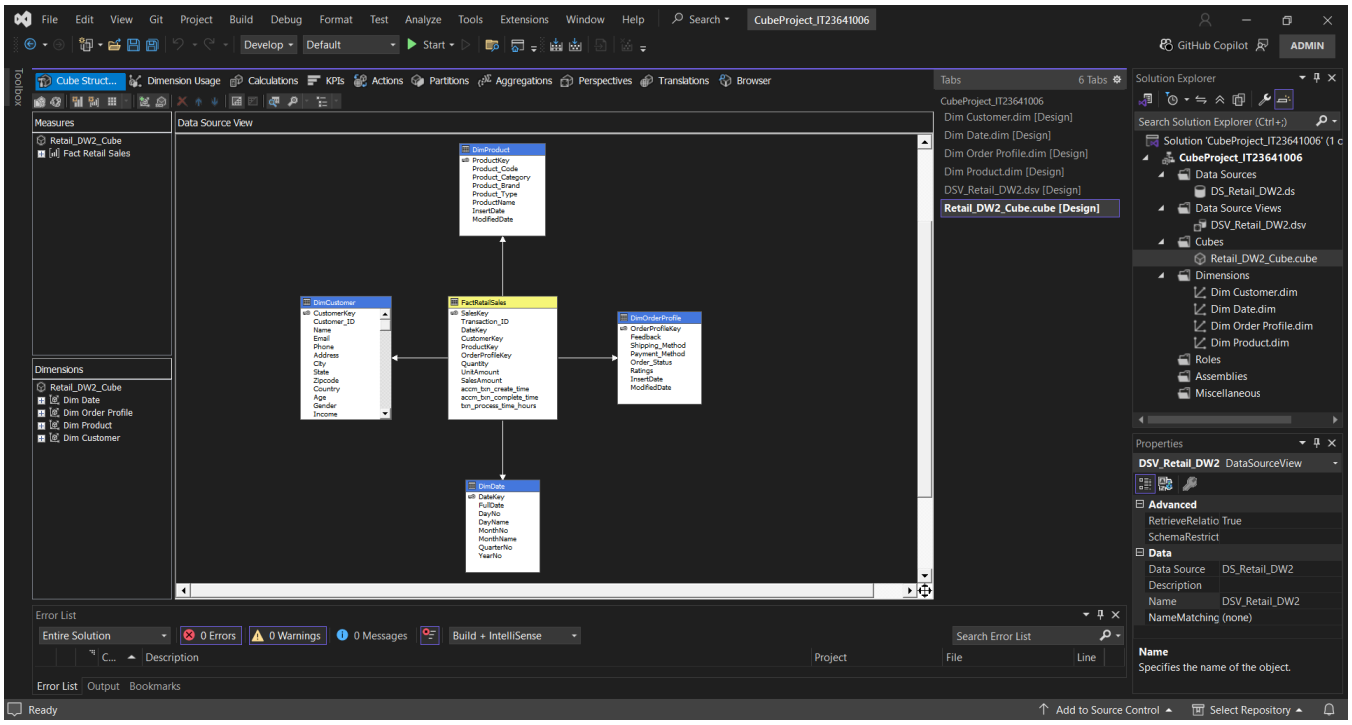
Calendar Hierarchy was implemented in Dim Date. The hierarchy uses YearNo > QuarterNo > MonthName > FullDate. This was later exposed in cube browsing and also supported to drill-down style analysis in the Excel and Power BI stages.



Product Hierarchy was implemented in Dim Product. The hierarchy uses Product Category > Product Brand > Product Name.



Cube structure and Data Source View after dimensions and cube objects.



3.3 Cube measures and dimension usage

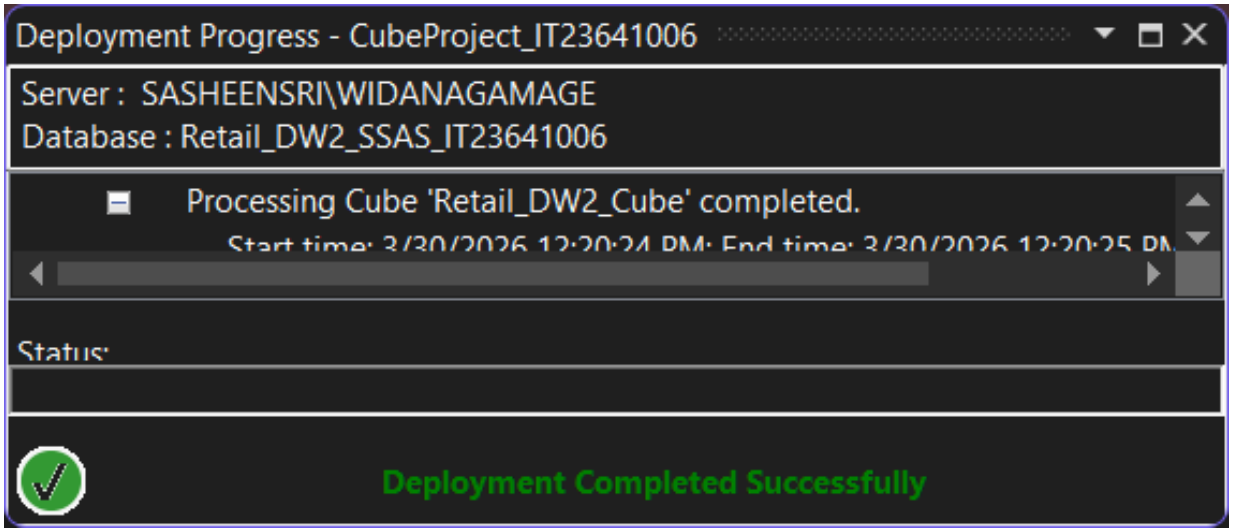
FactRetailSales was selected as the fact table and the measures selected for the cube were Quantity, Unit Amount, Sales Amount, Txn Process Time Hours and Fact Retail Sales Count.

Measures selected in the SSAS cube

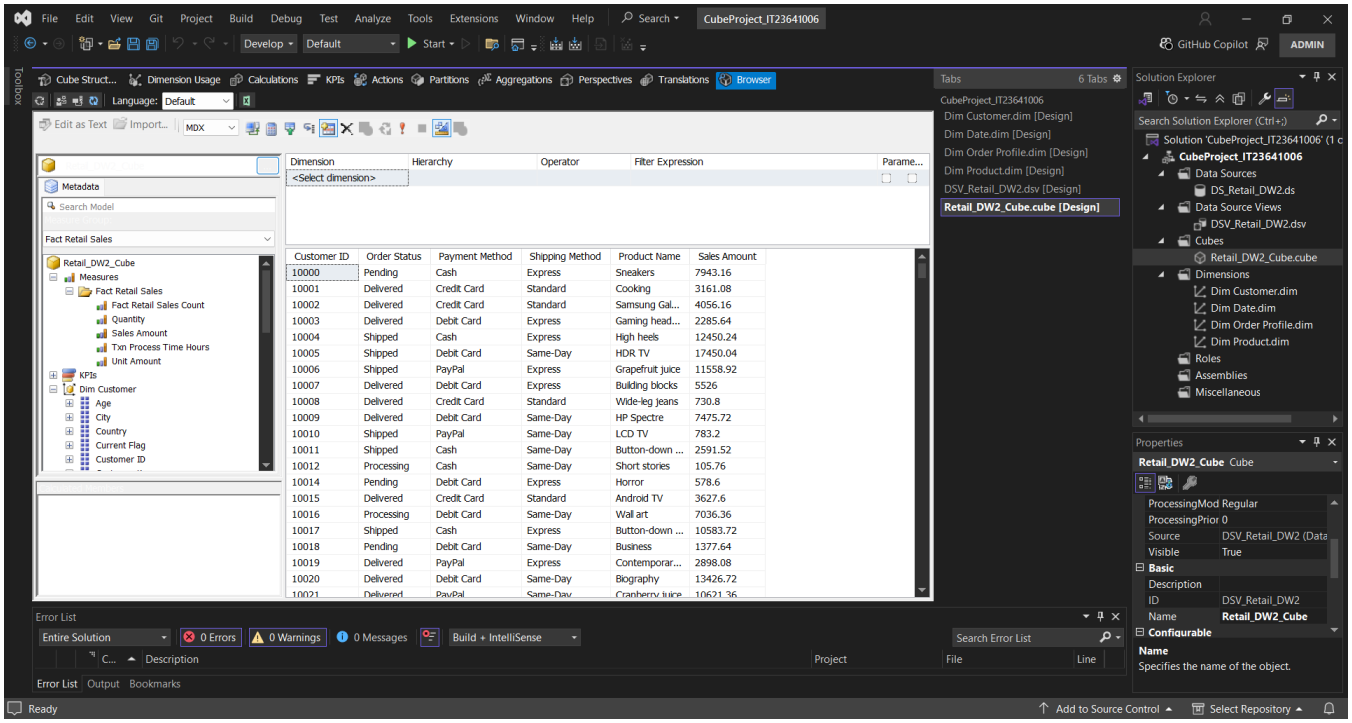
Cube measure	Purpose
Quantity	Units sold analysis
Unit Amount	Price / unit value analysis
Sales Amount	Total sales value analysis
Txn Process Time Hours	Operational efficiency / completion duration analysis
Fact Retail Sales Count	Number of transaction rows

3.4 Deployment, processing and browser validation

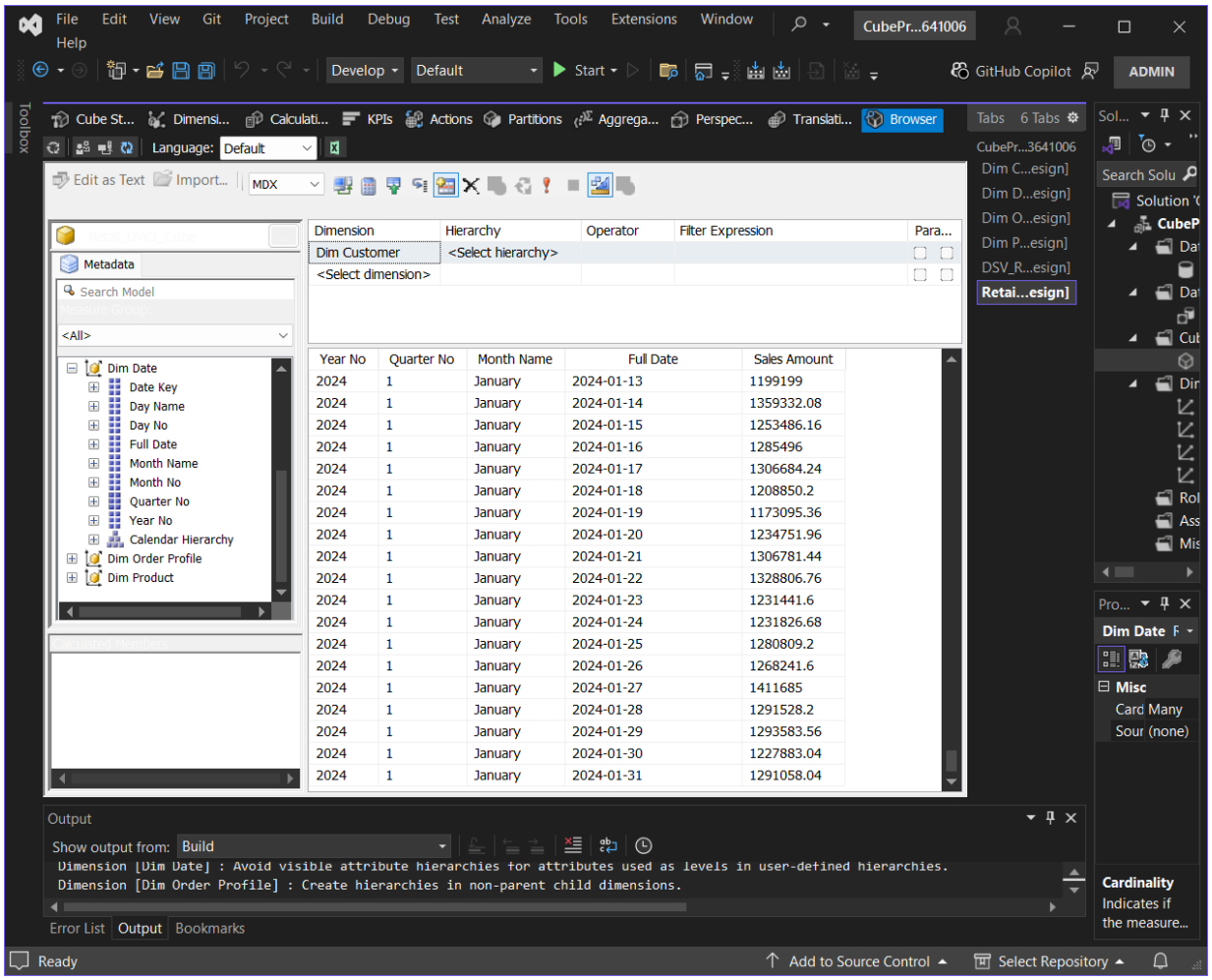
Successful deployment of Retail_DW2_Cube to Retail_DW2_SSAS_IT23641006.



SSAS cube browser showing Order Details.



SSAS cube browser showing Sales Amount by date hierarchy levels.



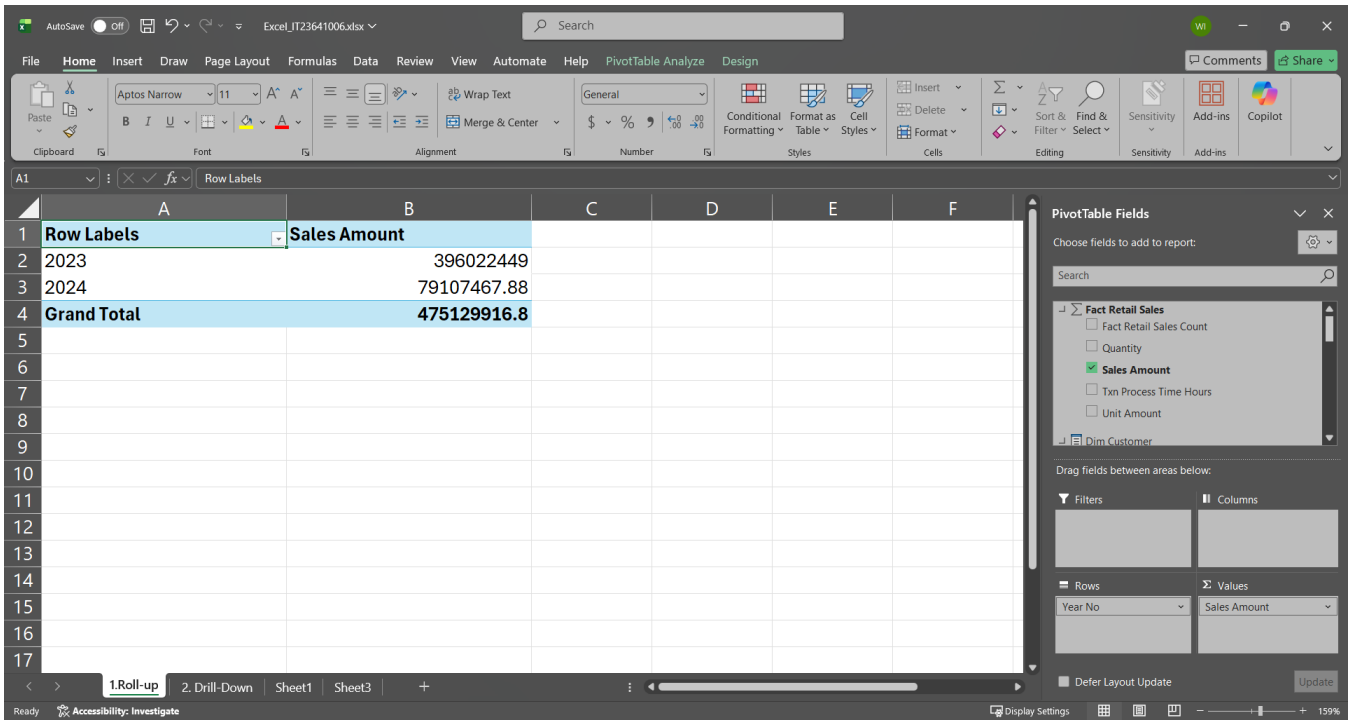
4. Demonstration of OLAP operations in Excel

4.1 Connecting Excel to the cube

Excel was connected to the deployed SSAS cube using the Data tab. The connection used the Analysis Services server SASHEENSRI\WIDANAGAMAGE, SSAS database Retail_DW2_SSAS_IT23641006, and cube Retail_DW2_Cube. The connection file was saved as Retail_DW2_Cube.odc and a PivotTable report was created in a new worksheet.

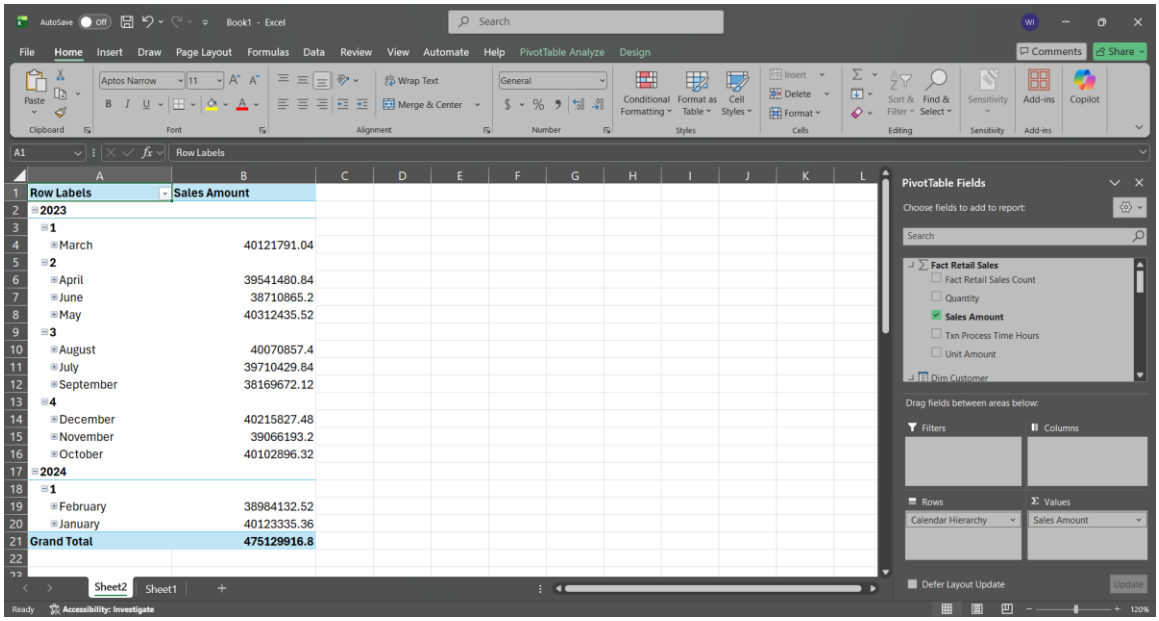
4.2 OLAP operation 1 - Roll-up

Roll-up was demonstrated by summarizing Sales Amount at aggregation(YearNo). This operation reduces the detail level and shows totals.



4.3 OLAP operation 2 - Drill-Down

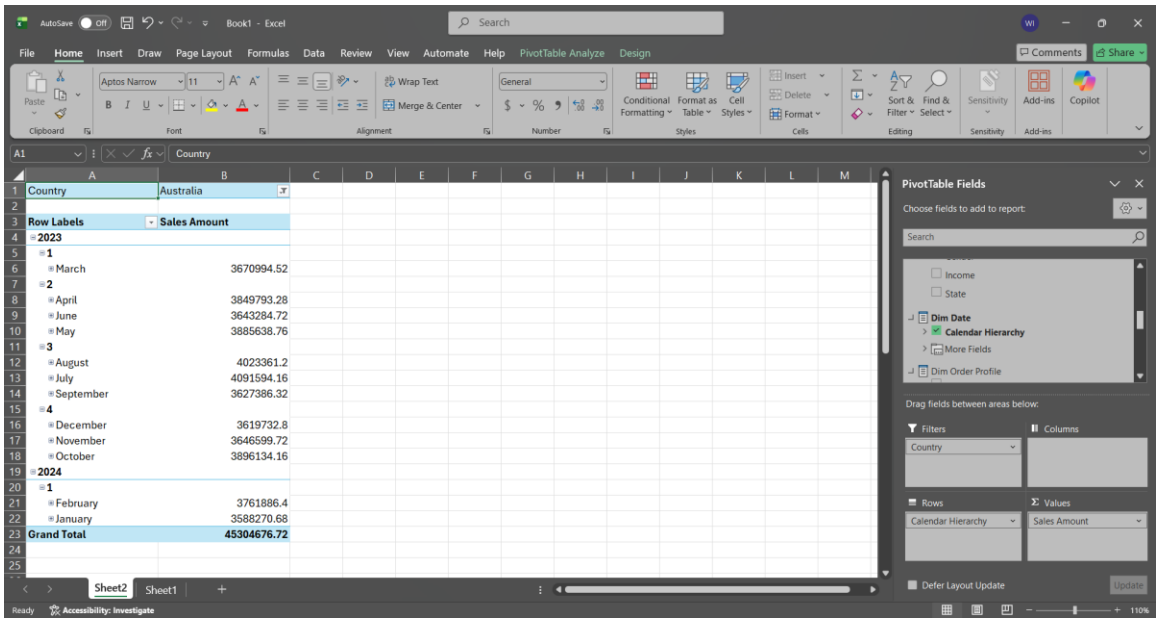
Drill-down was demonstrated by expanding the date hierarchy from YearNo to QuarterNo, MonthName and FullDate. This operation enables the user to start at a summary level and progressively move to lower levels of detail.



Row Labels	Sales Amount
2023	
1	
March	40121791.04
2	
April	39541480.84
June	38710865.2
May	40312435.52
3	
August	40070857.4
July	39710429.84
September	38169672.12
4	
December	40215827.48
November	39066193.2
October	40102896.32
2024	
1	
February	38984132.52
January	40123335.36
Grand Total	475129916.8

4.4 OLAP operation 3 – Slice

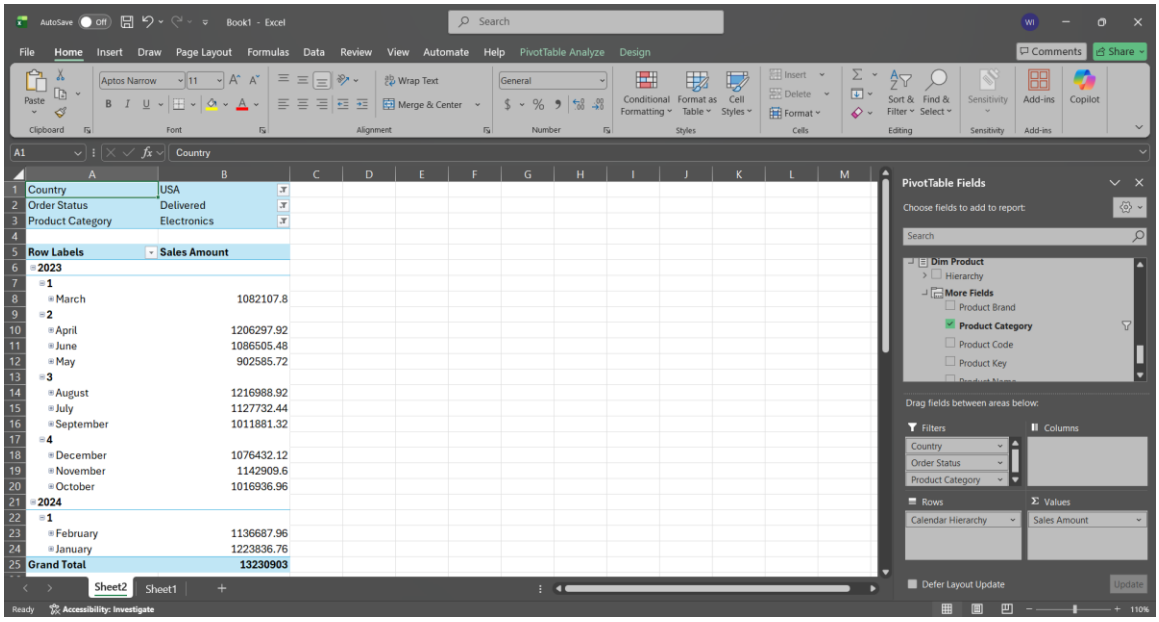
Slice was demonstrated by applying a filter on a single dimension value. This produces a single dimensional cut of the cube and allows focused analysis of a specific subset of the data.



Country	Australia
Row Labels	Sales Amount
2023	
1	
March	3670994.52
2	
April	3849793.28
June	3643284.72
May	3885638.76
3	
August	4023361.2
July	4091594.16
September	3627386.32
4	
December	3619732.8
November	3646599.72
October	3896134.16
2024	
1	
February	3761886.4
January	3588270.68
Grand Total	45304676.72

4.5 OLAP operation 4 – Dice

Dice was demonstrated by filtering multiple dimensions simultaneously. This creates a smaller sub-cube for detailed analysis under multiple conditions.

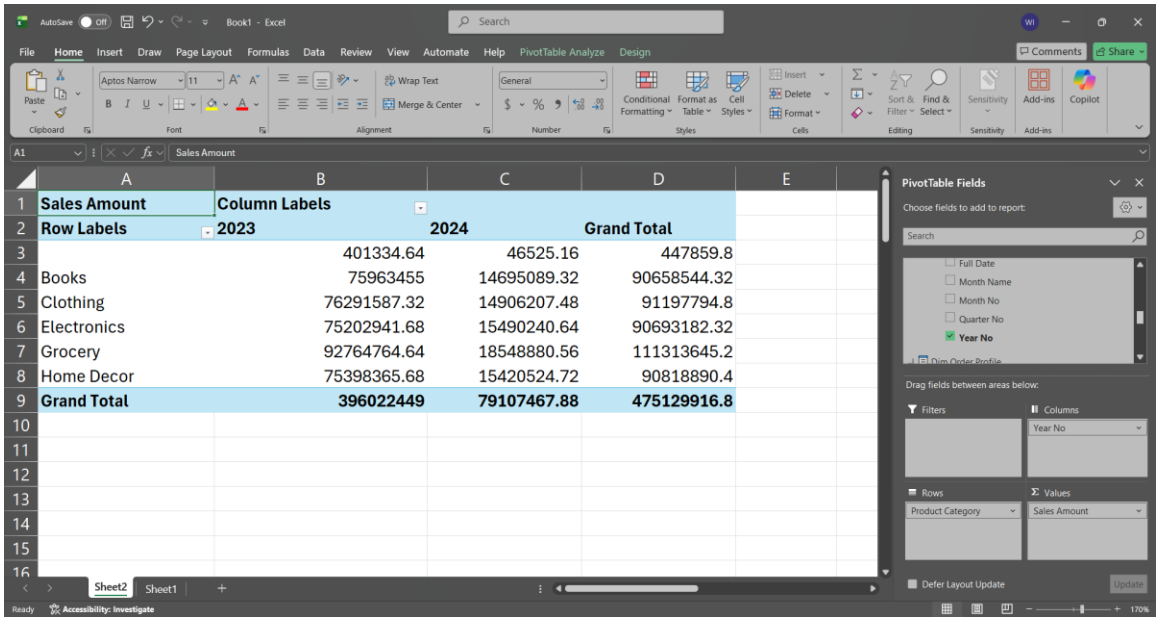


Country	USA
Order Status	Delivered
Product Category	Electronics

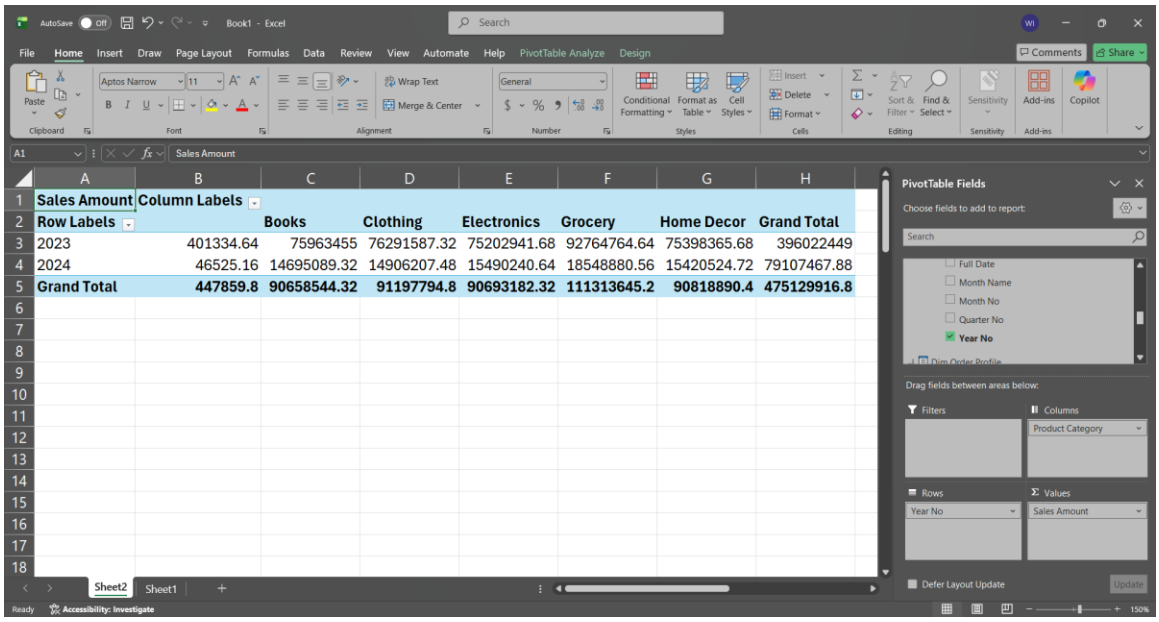
Row Labels	Sales Amount
2023	
1	
March	1082107.8
2	
April	1206297.92
June	1086505.48
May	902585.72
3	
August	1216988.92
July	1127732.44
September	1011881.32
4	
December	1076432.12
November	1142909.6
October	1016936.96
2024	
1	
February	1136687.96
January	1223836.76
Grand Total	13230903

4.6 OLAP operation 5 - Pivot

Pivot was demonstrated by rotating the analytical view, typically by moving Product_Category and YearNo between Rows and Columns. This preserves the same measure values but presents them from another analytical angle, making it easier to compare by product or by time.



Sales Amount	Column Labels		
Row Labels	2023	2024	Grand Total
	401334.64	46525.16	447859.8
Books	75963455	14695089.32	90658544.32
Clothing	76291587.32	14906207.48	91197794.8
Electronics	75202941.68	15490240.64	90693182.32
Grocery	92764764.64	18548880.56	111313645.2
Home Decor	75398365.68	15420524.72	90818890.4
Grand Total	396022449	79107467.88	475129916.8

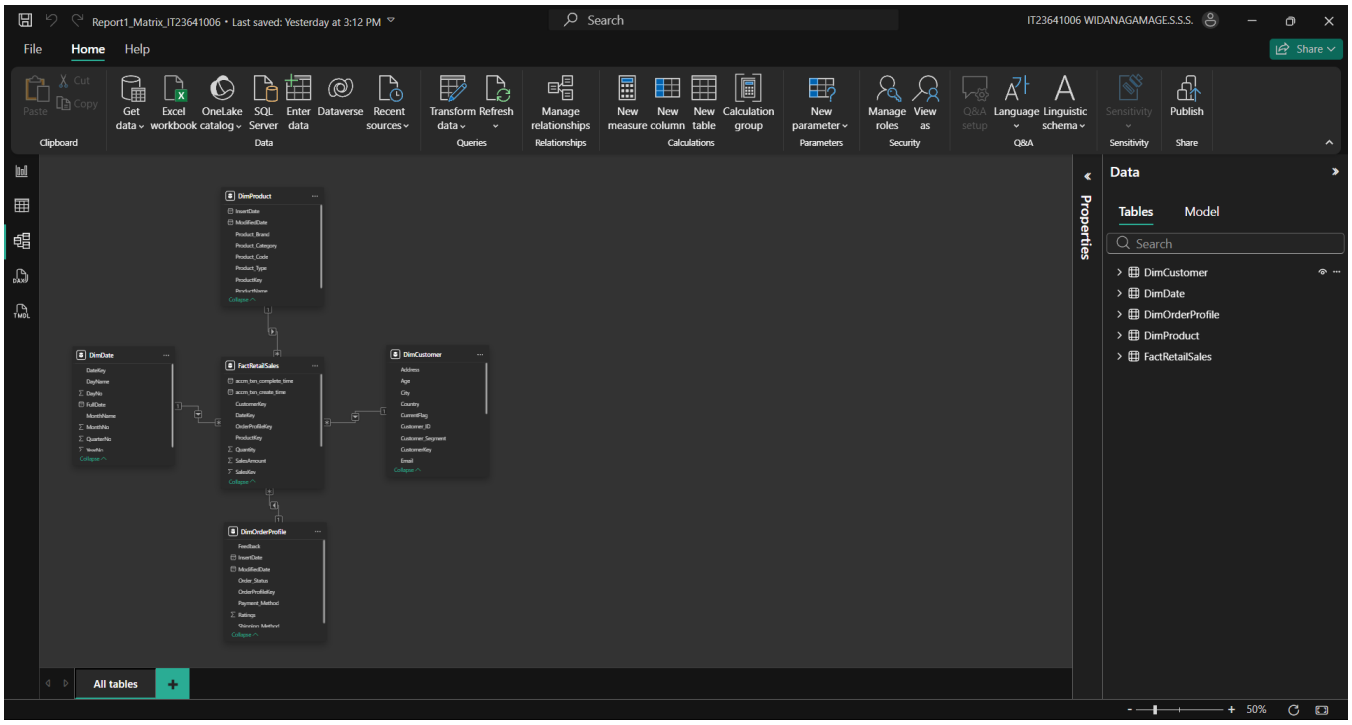


Sales Amount	Column Labels						
Row Labels		Books	Clothing	Electronics	Grocery	Home Decor	Grand Total
2023	401334.64	75963455	76291587.32	75202941.68	92764764.64	75398365.68	396022449
2024	46525.16	14695089.32	14906207.48	15490240.64	18548880.56	15420524.72	79107467.88
Grand Total	447859.8	90658544.32	91197794.8	90693182.32	111313645.2	90818890.4	475129916.8

5. Power BI Reports

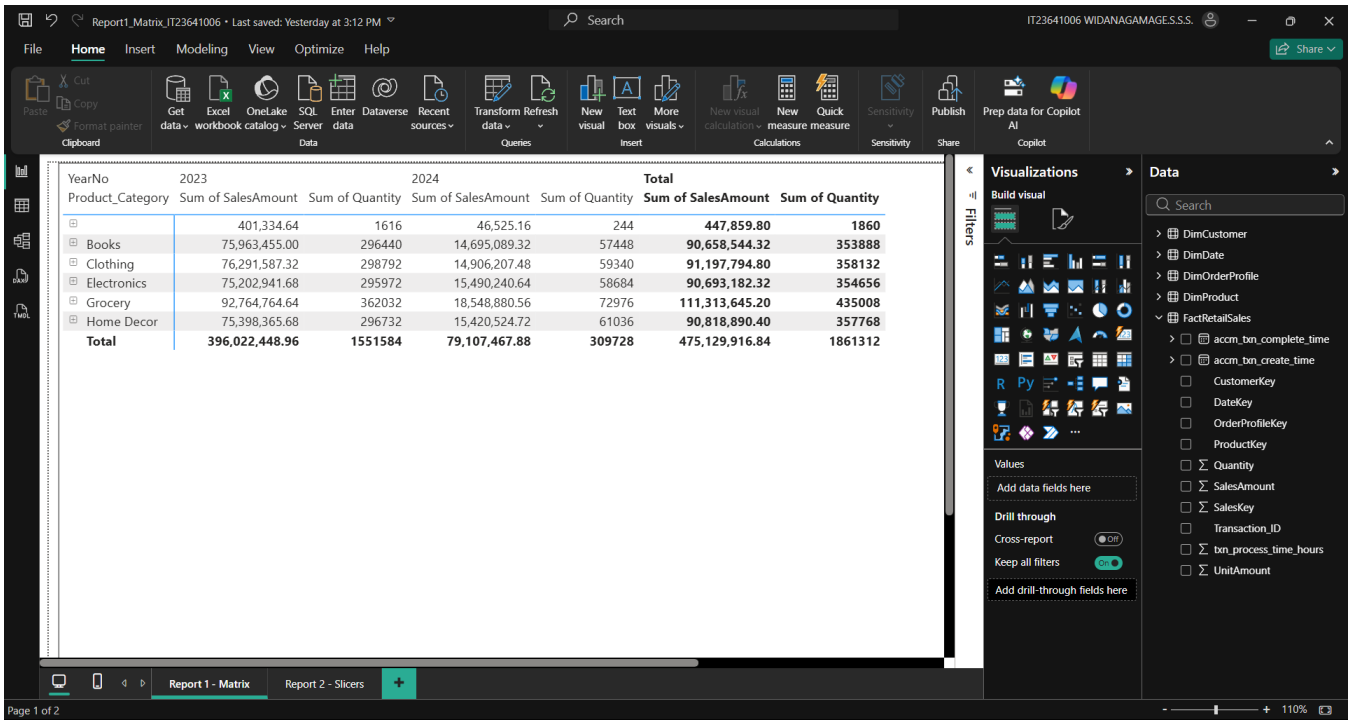
5.1 Data preparation and model relationships

Power BI imported DimCustomer, DimDate, DimOrderProfile, DimProduct and FactRetailSales from localhost\SASHEEN / Retail_DW2. The model view confirmed the correct star-schema relationships, with the dimension tables on the 1-side and FactRetailSales on the many-side.



5.2 Report 1 - Matrix visual

Report 1 used a Matrix visual to display detail with row and column groupings. Product_Category was placed on rows, YearNo was placed on columns. The measures SalesAmount and Quantity from FactRetailSales were then added to the Values section using sum aggregation.



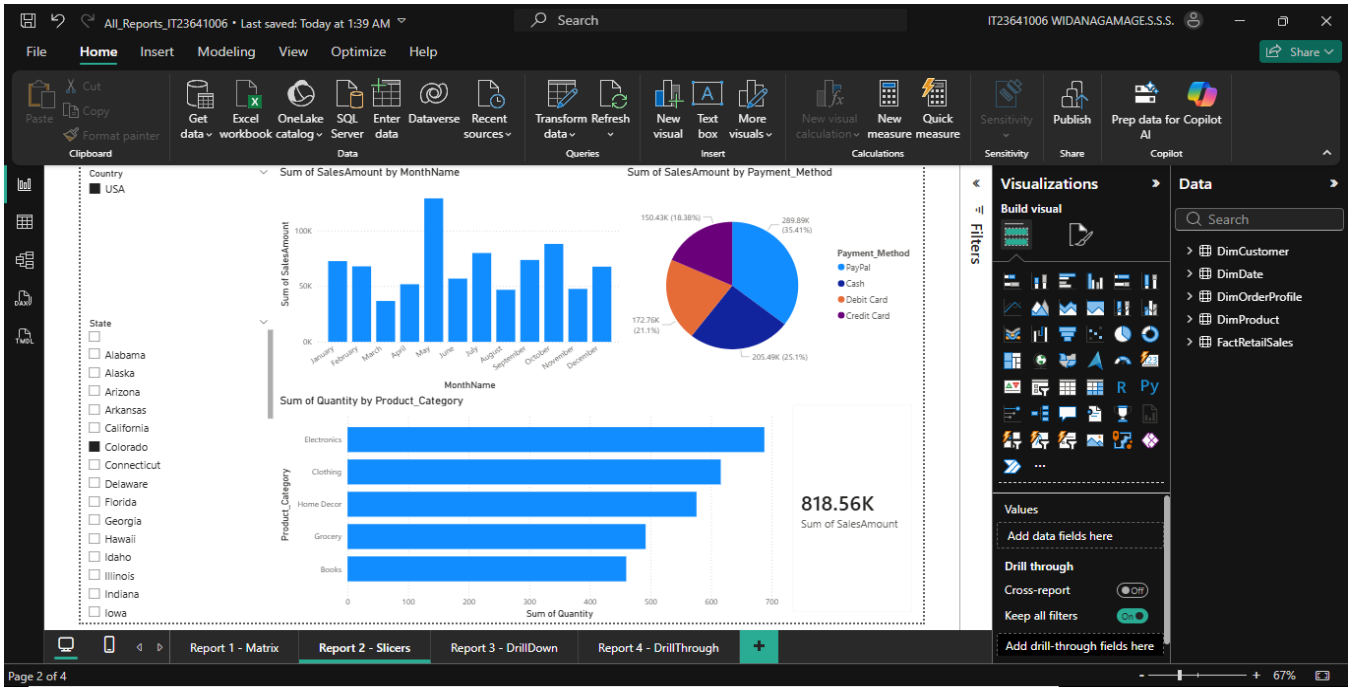
YearNo	2023		2024		Total	
Product_Category	Sum of SalesAmount	Sum of Quantity	Sum of SalesAmount	Sum of Quantity	**Sum of SalesAmount**	**Sum of Quantity**
	401,334.64	1616	46,525.16	244	**447,859.80**	**1860**
Books	75,963,455.00	296440	14,695,089.32	57448	**90,658,544.32**	**353888**
Clothing	76,291,587.32	298792	14,906,207.48	59340	**91,197,794.80**	**358132**
Electronics	75,202,941.68	295972	15,490,240.64	58684	**90,693,182.32**	**354656**
Grocery	92,764,764.64	362032	18,548,880.56	72976	**111,313,645.20**	**435008**
Home Decor	75,398,365.68	296732	15,420,524.72	61036	**90,818,890.40**	**357768**
Total	**396,022,448.96**	**1551584**	**79,107,467.88**	**309728**	**475,129,916.84**	**1861312**

5.3 Report 2 - Multiple slicers with cascading filter

Report 2 was developed as a slicer-driven analysis page using cascading geographic filters. The first slicer is **Country**, and the second slicer is **State**. When a country is selected, the State slicer is automatically restricted to only the valid states belonging to that country. The page contains four visuals,

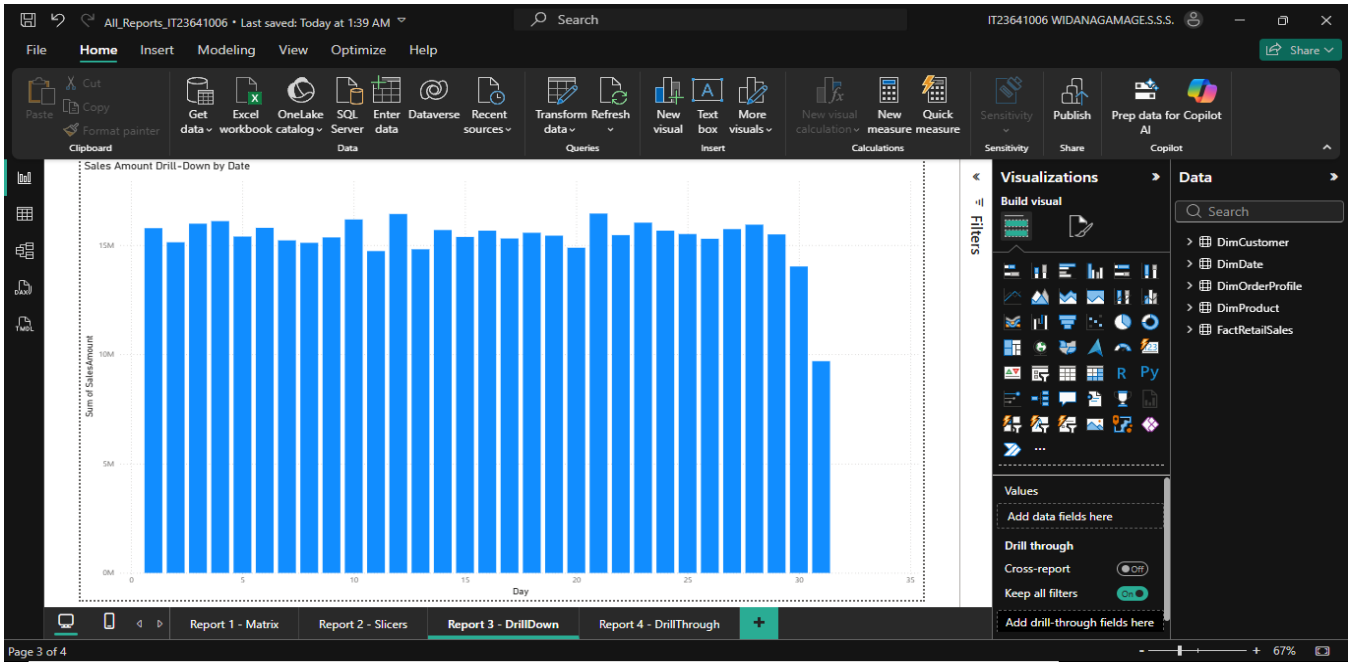
1. **Column chart** was created using MonthName from DimDate and Sum of SalesAmount from FactRetailSales to show how sales change across months.
2. **Pie chart** was created using Payment_Method from DimOrderProfile and Sum of SalesAmount to show the contribution of different payment methods to total sales.
3. **Horizontal bar chart** was created using Product_Category from DimProduct and Sum of Quantity from FactRetailSales to compare the quantity sold across product categories
4. **Card visual** displaying the overall Sum of SalesAmount

This report allows users to narrow analysis by location while simultaneously viewing sales trends over time, product-category quantity comparisons, and the total sales value.



5.4 Report 3 – Drill-down report

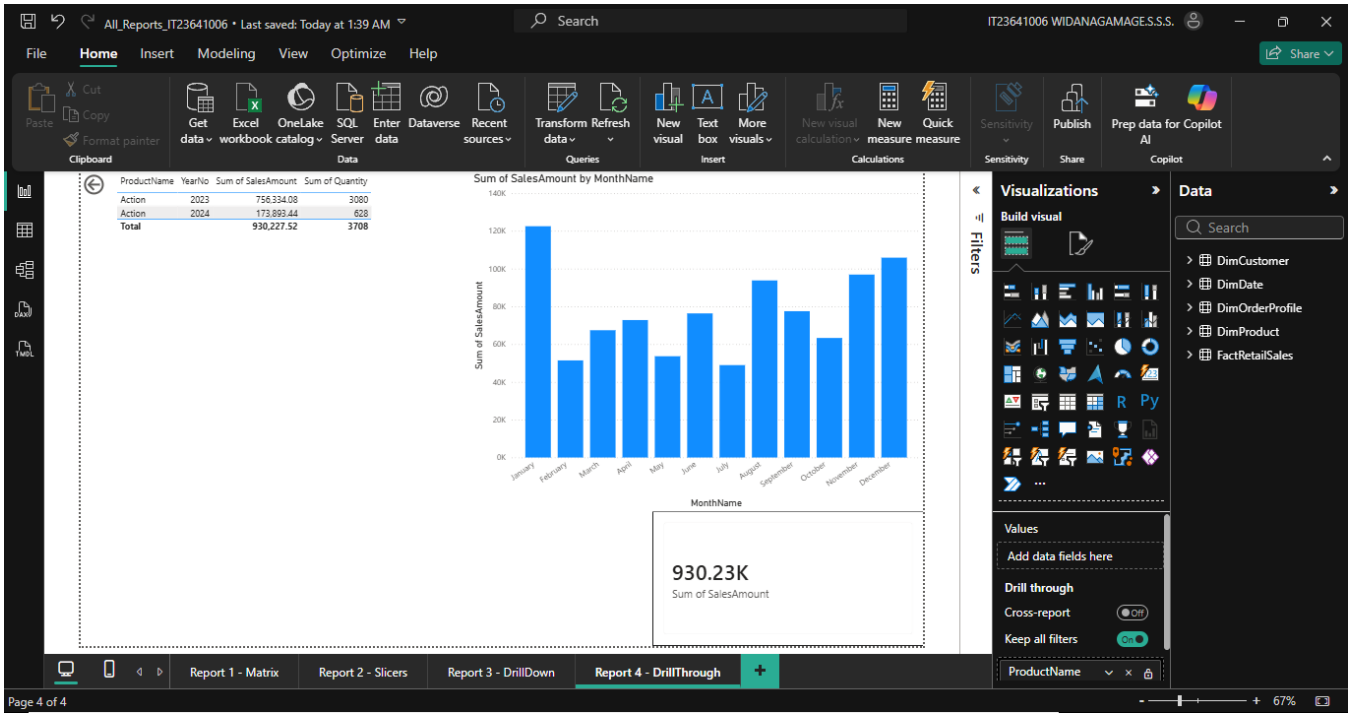
This report was created for **drill-down analysis** using a date hierarchy. A clustered column chart was inserted, and fields from DimDate were added in hierarchical order: YearNo, QuarterNo, MonthName, and FullDate. SalesAmount was used as the value field with sum aggregation.



5.5 Report 4 – Drill-through report

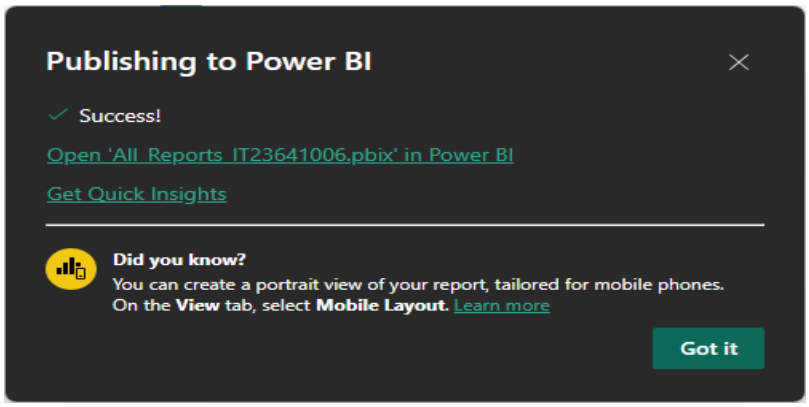
This was designed for **drill-through functionality**. To do this, a dedicated detail page was created. On this page, ProductName from DimProduct was added to the **Drill-through filters** area. When user right-clicks a product in another report and selects drill-through, Power BI opens the detail page and automatically filters it to selected product.

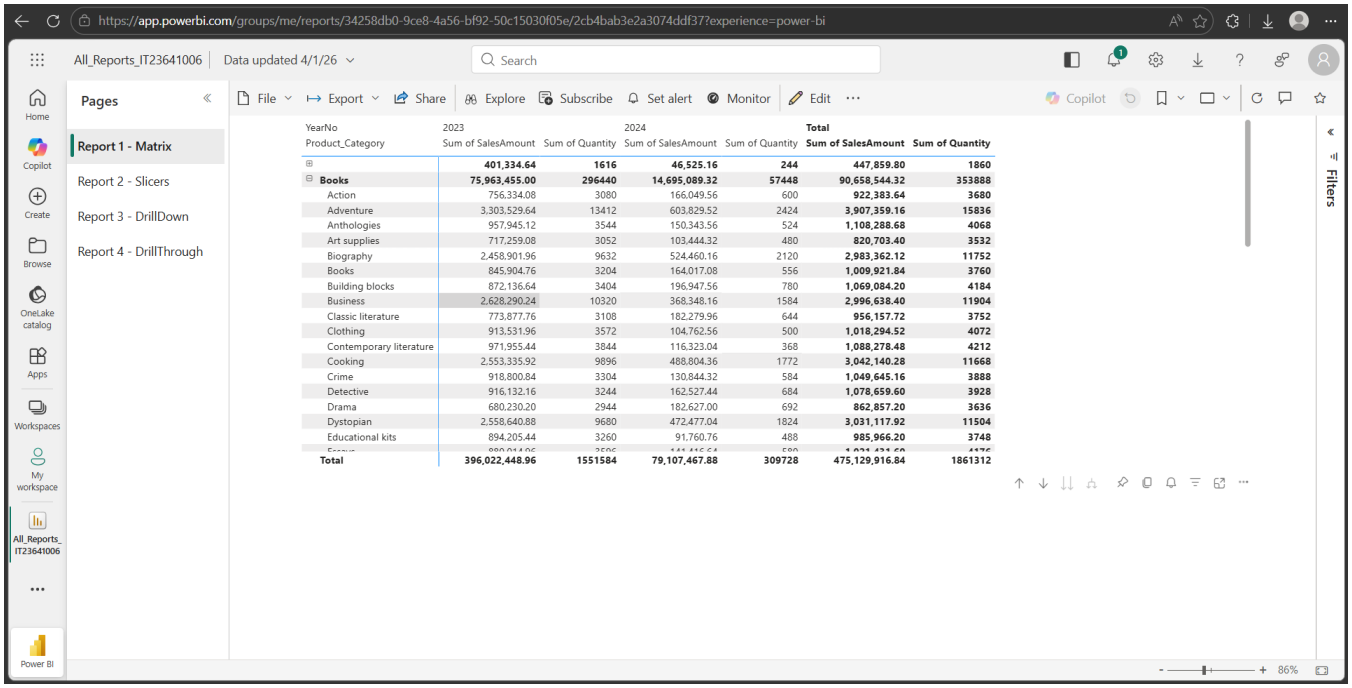
The drill-through page contains a **table visual**, a **card visual**, and a **monthly sales chart**. The table was used to show detailed information related to the selected product, the card was used to summary value, and the chart was used to show sales over time for product.



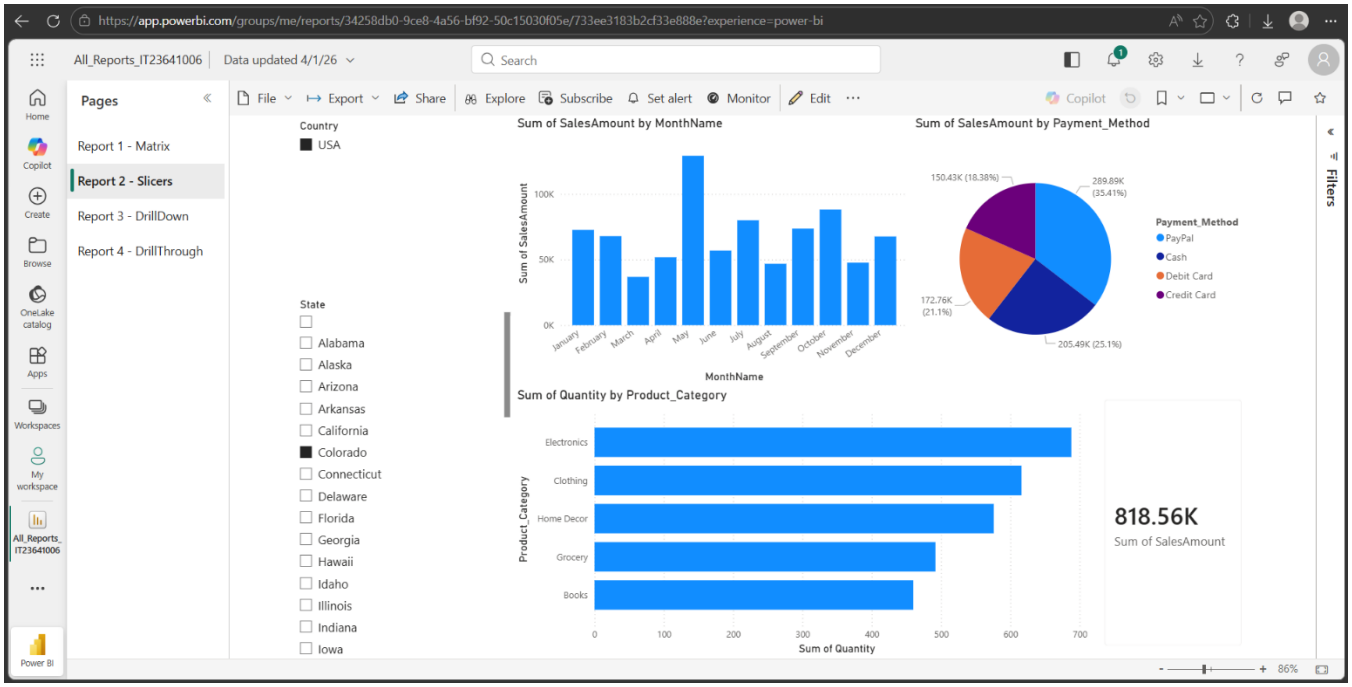
5.6 Publishing to Power BI Service

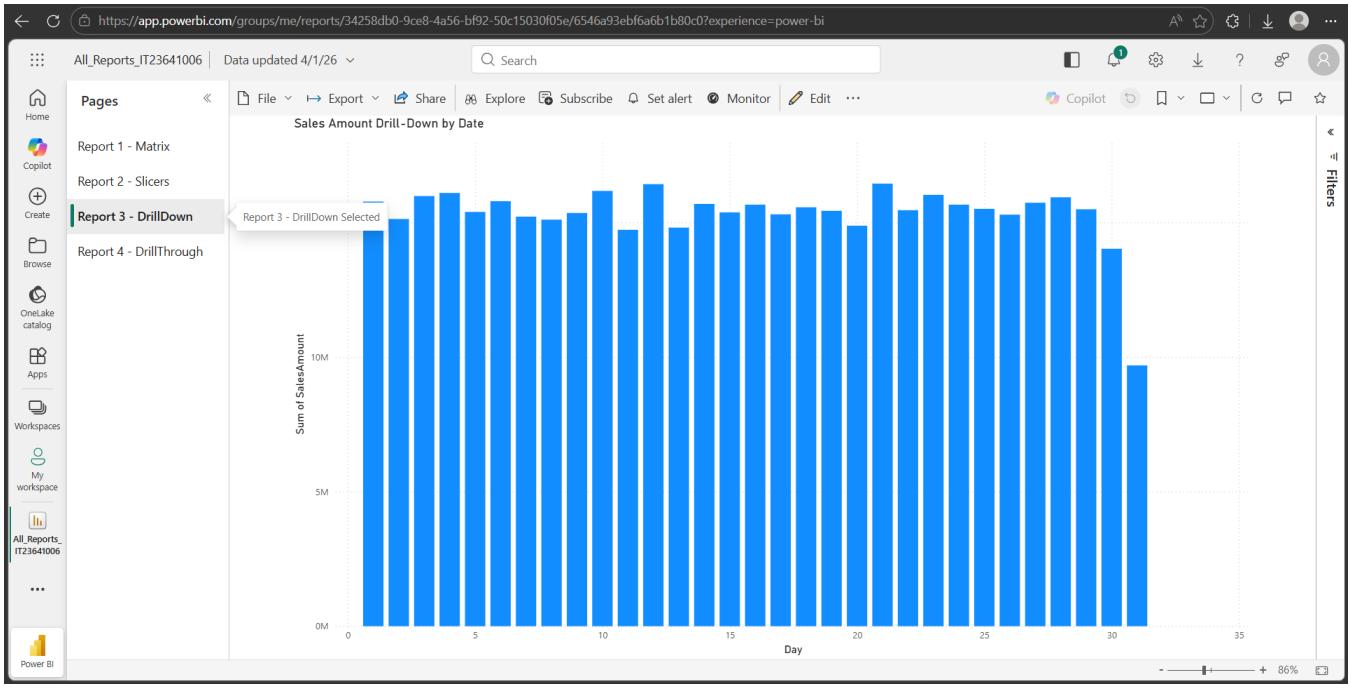
After all four report pages were created and tested in Power BI Desktop, the report file was published to Power BI Service using the Publish option.





YearNo	2023		2024		Total	
Product_Category	Sum of SalesAmount	Sum of Quantity	Sum of SalesAmount	Sum of Quantity	Sum of SalesAmount	Sum of Quantity
	401,334.64	1616	46,525.16	244	447,859.80	1860
Books	**75,963,455.00**	**296440**	**14,695,089.32**	**57448**	**90,658,544.32**	**353888**
Action	756,334.08	3080	166,049.56	600	922,383.64	3680
Adventure	3,303,529.64	13412	603,829.52	2424	3,907,359.16	15836
Anthologies	957,945.12	3544	150,343.56	524	1,108,288.68	4068
Art supplies	717,259.08	3052	103,444.32	480	820,703.40	3532
Biography	2,458,901.96	9632	524,460.16	2120	2,983,362.12	11752
Books	845,904.76	3204	164,017.08	556	1,009,921.84	3760
Building blocks	872,136.64	3404	196,947.56	780	1,069,084.20	4184
Business	2,628,290.24	10320	368,348.16	1584	2,996,638.40	11904
Classic literature	773,877.76	3108	182,279.96	644	956,157.72	3752
Clothing	913,531.96	3572	104,762.56	500	1,018,294.52	4072
Contemporary literature	971,955.44	3844	116,323.04	368	1,088,278.48	4212
Cooking	2,553,335.92	9896	488,804.36	1772	3,042,140.28	11668
Crime	918,800.84	3304	130,844.32	584	1,049,645.16	3888
Detective	916,132.16	3244	162,527.44	684	1,078,659.60	3928
Drama	680,230.20	2944	182,627.00	692	862,857.20	3636
Dystopian	2,558,640.88	9680	472,477.04	1824	3,031,117.92	11504
Educational kits	894,205.44	3260	91,760.76	488	985,966.20	3748
Total	**396,022,448.96**	**1551584**	**79,107,467.88**	**309728**	**475,129,916.84**	**1861312**

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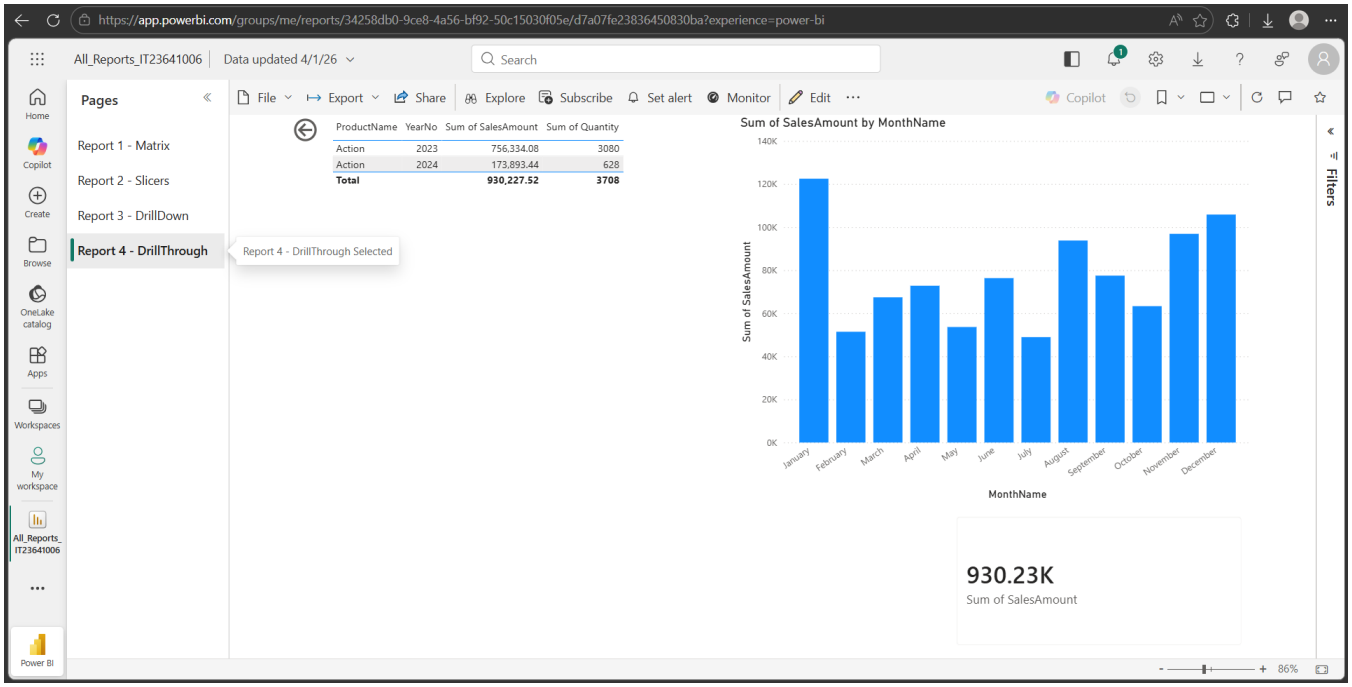
Data updated 4/1/26
Search
Pages
Report 1 - Matrix
Report 2 - Slicers
Report 3 - DrillDown
Report 4 - DrillThrough
Report 3 - DrillDown Selected
File
Export
Share
Explore
Subscribe
Set alert
Monitor
Edit
Copilot
Sales Amount Drill-Down by Date
Sum of SalesAmount
10M
5M
0M
0
5
10
15
20
25
30
35
Day
Filters
86%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All_Reports_IT23641006
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Search
Pages
Report 1 - Matrix
Report 2 - Slicers
Report 3 - DrillDown
Report 4 - DrillThrough
Report 4 - DrillThrough Selected
File
Export
Share
Explore
Subscribe
Set alert
Monitor
Edit
Copilot
ProductName YearNo Sum of SalesAmount Sum of Quantity
Action 2023 756,334.08 3080
Action 2024 173,893.44 628
Total 930,227.52 3708
Sum of SalesAmount by MonthName
Sum of SalesAmount
140K
120K
100K
80K
60K
40K
20K
0K
January
February
March
April
May
June
July
August
September
October
November
December
MonthName
930.23K
Sum of SalesAmount
Filters
86%